Metabolites	Examples
Primary	
Enzymes	protease, lipase, peroxidase
Amino acid	proline, leucine
Organic acid	acetic acid, lactic acid
Vitamins	A, B, C
Secondary	
Pigments	carotenoids, anthocyanins
Alkaloids	morphine, codeine
Essential oil	lemon grass oil, rose oil
Toxins	abrin, ricin
Lectins	concanavalin A
Drugs	vinblastin, curcumin
Polymeric substances	rubber, gums, cellulose

8.2.1 Organic Molecules

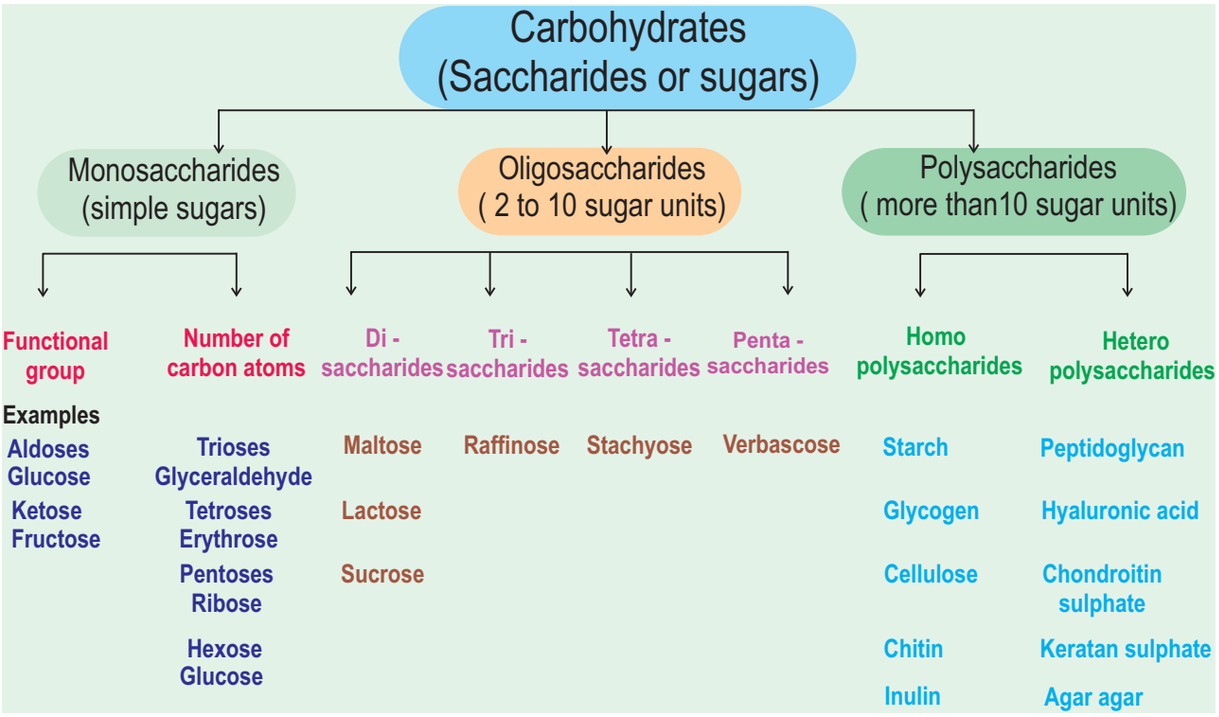
Organic molecules may be small and simple. These simple molecules assemble and form large and complex molecules called **macromolecules**. These include four main classes – carbohydrates, lipids, proteins and nucleic acids. All macromolecules except lipids are formed by the process of polymerisation, a process in which repeating subunits termed monomers are bound into chains of different lengths. These chains of monomers are called **polymers**.

8.3 Carbohydrates

Carbohydrates are organic compounds made of carbon and water. Thus one molecule of water combines with a carbon atom to form CH_2O and is repeated several (n) times to form $(CH_2O)_n$ where n is an integer ranging from 3–7. These are also called as **saccharides**. The common term sugar refers to a simple carbohydrate such as a monosaccharide or disaccharide that tastes sweet are soluble in water (Figure 8.7).

8.3.1 Monosaccharides – The Simple Sugars

Monosaccharides are relatively small molecules constituting single sugar unit.



Classification of Carbohydrates

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